MIL-B-40042E
22 January 1988
SUPERSEDING
MIL-B-40042D
6 September 1974

MILITARY SPECIFICATION

BOX, MATCH, WATERPROOF

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a plastic waterproof matchbox with integrally mounted striking disc.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-349	-	Plastic Molding and Extrusion Material, Cellulose Acetate Butyrate
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-T-45	-	Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8465

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MILITARY

MIL-L-35078 - Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For

STANDARDS

FEDERAL

FED-STD-406 - Plastics: Methods of Testing
FED-STD-595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

* 2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

8-2-10 - Box, Match, Waterproof

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 787 - Ethyl Cellulose Molding and Extrusion Compounds

D 2000 - Elastomeric Materials for Automotive Applications

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Material. (See 6.4)

3.2.1 Molding compound. The compound used in molding the box and cap shall be cellulose acetate butyrate conforming to type III, class MH of L-P-349 or ethyl cellulose conforming to type II, grade 7 of ASTM D 787.

* 3.2.2 Sealing gasket. The sealing gasket shall be made from rubber material meeting the requirements of type, class, grade and suffix letters M4BA610A14Z of ASTM D 2000. Suffix letter "Z" indicates that there is a special requirement that the gasket material, in conjunction with the adhesive (3.2.4), striking disc (3.2.3), and the cap compound (3.2.1) shall be of a type which will permit the adhesion requirements of 3.4.2 to be met. The diameter of the gasket shall not be larger than the inside diameter of the cap and shall be not more than 1/64 inch smaller that the inside diameter of the cap.

3.2.3 Striking disc. The striking disc shall be commercial waterproof abrasive cloth, approximately 120 grit.

3.2.4 Adhesive. The adhesive or adhesives used for bonding shall bond the sealing gasket to the cap and the striking disc to the sealing gasket. The adhesive or adhesives shall be of a type and quality to meet the applicable performance requirements specified in 3.4.1, 3.4.2, and 3.4.3.

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3.3 Design and construction. The matchbox shall conform to the details of design, construction, and dimensions shown on Drawing 8-2-10.

3.4 Performance.

3.4.1 Waterproofness. The finished matchbox shall exhibit no penetration of water when tested as specified in 4.4.4.

3.4.2 Adhesion of sealing gasket and striking disc. When tested as specified in 4.4.4, the sealing gasket shall not separate either partially or completely from the matchbox cap, and the striking disc shall not separate either partially or completely from the sealing gasket.

3.4.3 Dimensional stability. When tested as specified in 4.4.4, the matchbox shall not exhibit visual evidence of warpage and shall be free from cracks, swelling, or distortion. The cap shall be capable of moving freely on its threads. The striking disc and sealing gasket shall not have loosened or changed position.

3.5 Color and luster. The color and luster of the matchbox shall approximate color No. 14087 of FED-STD-595.

3.6 Instruction sheet. Each matchbox shall be furnished with an instruction sheet inside. The instruction sheet shall consist of a piece of white paper 3 inches by 3 inches printed in boldfaced type with the following legend:

"PLACE MATCHES IN MATCHBOX WITH THE HEAD OF THE MATCH TOWARD THE BOTTOM"

3.7 Marking for identification. Each matchbox shall have the letters "US" and the manufacturer's name or brand name embossed on the bottom as shown in Drawing 8-2-10.

3.8 Workmanship. The matchbox shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsiblity for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in table I and tested for the characteristics specified in 4.4.4. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.2 End item visual examination. The end items shall be examined for the defects listed in table I. The lot size shall be expressed in units of matchboxes. The sample unit shall be one finished matchbox. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5.

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TABLE I. End item visual defects

Examine	Defect
Material:	
Molded plastic	Color and luster not specified Discolored, faded, resin-streak or mottled Wrinkle, pimple, or crater Crazing, pitted, rough, or porous surface Surface scaly, hazy, chalky, or burned Any dirt in molding material Any deep scratch or gouge mark on the exterior of the box
NOTE:	Fine scratches may be disregarded.
Sealing gasket	Gasket not in one piece Crack, extending through body Cut, tear, or hole Bruise, loose skin, spongy, scar, void, or otherwise defective
Striking disc	Torn, area of grit missing, or otherwise damaged
Design and construction	Any part or component not fabricated of applicable referenced material (i.e., striking disc is not waterproof abrasive cloth) Varies from design, contour, or construction shown on specified drawing Any characteristic not in accordance with specified requirements (i.e., diameter of gasket is larger than the inside diameter of the cap or diameter of gasket is more than 1/64 inch smaller than the inside diameter of the cap)
Workmanship and assembly	Threads missing or not type required Threads broken, stripped, fractured, or otherwise impaired Flash on threads of box or cap Cap does not move freely on its threads Cap cannot be screwed on matchbox without pressure Rough, sharp, or broken edge on cap or body

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TABLE I. End item visual defects - Continued

Examine	Defect
Workmanship and assembly - continued	Striking disc or sealing gasket detached Adhesive extends beyond rim of striking disc Adhesive tacky Cap cannot be screwed on to give tight fit Sealing gasket missing Sealing gasket loose or changes position Sealing gasket bulged out from too tight a fit Striking disc missing or loose Striking disc not positioned in center of gasket Striking disc not positioned with the abrasive side showing Any depression or void (other than superficial) on interior bottom of the matchbox Surface and curves not blended smoothly Body or cap warped, twisted, deformed, bent out of shape, split, or otherwise impaired
Instruction sheet	Missing or not as specified Incomplete or illegible
Marking	Missing, incomplete, illegible, misspelled, not specified type, not in specified location, or not applied in the specified manner

* 4.4.3 End item dimensional examination. The end items shall be examined for for conformance to the dimensions, clearance between bottom of cap and rim of body, and thread requirements annotated with an asterisk (*) on Drawing 8-2-10. Any failure to meet any of the requirements shall be classified as a defect. The lot size shall be expressed in units of matchboxes. The sample unit shall be one matchbox. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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4.4.4 End item testing. The end items shall be tested for the characteristics listed in table II. The lot size shall be expressed in units of matchboxes. The sample unit shall be four finished matchboxes. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 1.5.

TABLE II. End item tests

Characteristic	Requirement paragraph	Test method	No. determinations per sample unit	Results reported as
Waterproofness of matchbox	3.4.1	4.5.1	1	Pass or fail
Adhesion of sealing gasket:				
As is	3.4.2	4.5.2	1	Pass or fail
Oven aged	3.4.2	4.5.2	1	Pass or fail
After water immersion	3.4.2	4.5.2	1	Pass or fail
Adhesion of striking disc:				
As is	3.4.2	4.5.2	1	Pass or fail
Oven aged	3.4.2	4.5.2	1	Pass or fail
After water immersion	3.4.2	4.5.2	1	Pass or fail
Dimensional stability:				
After exposure at 140°F	3.4.3	4.5.3	1	Pass or fail
After exposure at -40°F	3.4.3	4.5.3	1	Pass or fail
Drop test	3.4.3	4.5.3	1	Pass or fail

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* 4.4.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number of unit containers is more or less than specified Number of matchboxes per unit container is more or less than specified 1/

1/ For this defect one unit container shall be examined.

4.4.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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4.5 Methods of inspection.

4.5.1 Waterproofness of matchbox. Condition the completely assembled matchbox for 24 hours at 73.5° ± 2°F and 50 percent ± 4 percent relative humidity. Put six wooden matches (nonsafety type) in the box head down and screw the cap on tightly by hand. Submerge the matchbox for 24 hours under 2 feet of tap water at room temperature (70° to 77°F). At the end of this period, remove the matchbox from the water and dry with an absorbent material to remove surface moisture. Open the box and strike each match against the striking disc not more than ten times to test for ignitability. Failure of two or more matches to ignite shall constitute failure of the waterproofness of matchbox test.

4.5.2 Adhesion of sealing gasket and stiking disc. A matchbox cap complete with sealing gasket and striking disc shall be subjected to four consecutive impacts of 12 inch-pounds each, as follows: Place the cap open side down on a smooth flat steel base at least 4 inches by 8 inches by 3/4 inch thick. Center the cap under a vertical tube which guides a dropping steel rod. The rod shall be 1 inch diameter and of a length to weigh 2.0 pounds, with the bottom end of the rod flat and squared to the centerline. The guide tube shall be not larger than 1 1/16 inch in diameter and shall permit an unobstructed free fall of the rod. Place a 1-inch diameter steel bearing ball on the top surface of the cap and position the guide tube so as to keep the ball in a central position on the cap. Raise the rod a height of 6 inches above the top of the ball and allow it to drop. The cap shall be impacted four times in this manner. Examine the gasket and disc visually for partial or complete separation from their positions. Test a separate matchbox cap complete with gasket and disc in the same way after being subjected to oven aging at 140°F ± 2°F for 24 hours and conditioned for 4 hours mimimum under standard conditions. Separation of gasket or disc in either the oven-aged or non-oven-aged specimen will constitute failure in this test. Immerse the matchbox cap complete with sealing gasket and striking disc for 24 hours in tap water at 70° to 77°F. At the end of this period, remove the cap and examine the gasket and disc visually for separation in whole or in part from their positions.

4.5.3 Dimensional stability. Condition the completely assembled matchbox for 24 hours at 73.5°F ± 2°F and 50 percent ± 4 percent relative humidity. Subject the matchbox to the accelerated service test in accordance with Method 6011 of FED-STD-406, except that the following exposure cycle periods shall be used:

Period of exposure (hours)	Temperature (°F)	Relative humidity (%)
24	140	85-89
24	-40	---

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Examine the matchbox immediately at the end of each 24-hour period for visual evidence of warpage, cracks, loss in luster, distortion, or swelling. After exposure to -40°F, drop the matchbox on a concrete floor or similarly hard surface from a minimum height of 3 feet. The striking disc and sealing gasket shall not become detached, loosened, or cracked. It is not required that percentage changes in weight or dimensions be reported.

5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. One hundred matchboxes, each with their caps turned finger tight, shall be unit packed in a fiberboard box conforming to style RSC, type CF (variety SW), class domestic, grade 200 of PPP-B-636. The matchboxes shall be packed upright ten in length, ten in width, and one in depth within each fiberboard box. Inside dimensions of the box shall be 12 inches in length, 12 inches in width, and 3 inches in depth. Box closure shall be secured with 2 inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

* 5.1.2 Commercial. Matchboxes shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. Six hundred matchboxes, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed flat, two in length, one in width, and three in depth within each shipping container. Inside dimensions of each shipping container shall be approximately 25 1/4 inches in length, 12 3/4 inches in width, and 11 1/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V and reinforced as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

* 5.2.2 Level B packing. Six hundred matchboxes, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, two in length, one in width, and three in depth within each shipping container. Inside dimensions of each shipping

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container shall be approximately 25 1/4 inches in length, 12 3/4 inches in width, and 11 1/4 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

* 5.2.2.1 Weather-resistant shipping container. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Matchboxes, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), matchboxes, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 91 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The matchbox (with contents) is intended to be included in military personnel survival kits, with no limitations as to climatic conditions.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1).
c. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
d. Type and class of unit load required (see 5.2.1).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

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* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

* 6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.5 Subject term (key word) listing.

Box
Match
Survival kits

6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - MC
Air Force - 99

Review activity:

Army - GL
Navy - MS
Air Force - 82
DLA - CT

User activity:

Air Force - 45

Preparing activity:

Army - GL

Project No. 8465-0979

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-40042E	Box, Match, Waterproof

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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NATICK OP, 1 Apr 84

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